

Proell Governors

by

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Gravity Loaded or Dead weight Governor:

- Proell Governor is a different type of governor in which the **fly balls are connected to the spindle using an extended arm**.
- Proell governor also has central weight which increases the speed of rotation.
- Proell Governor functions much more accurately than other governors and maintains a constant function without any fluctuation.

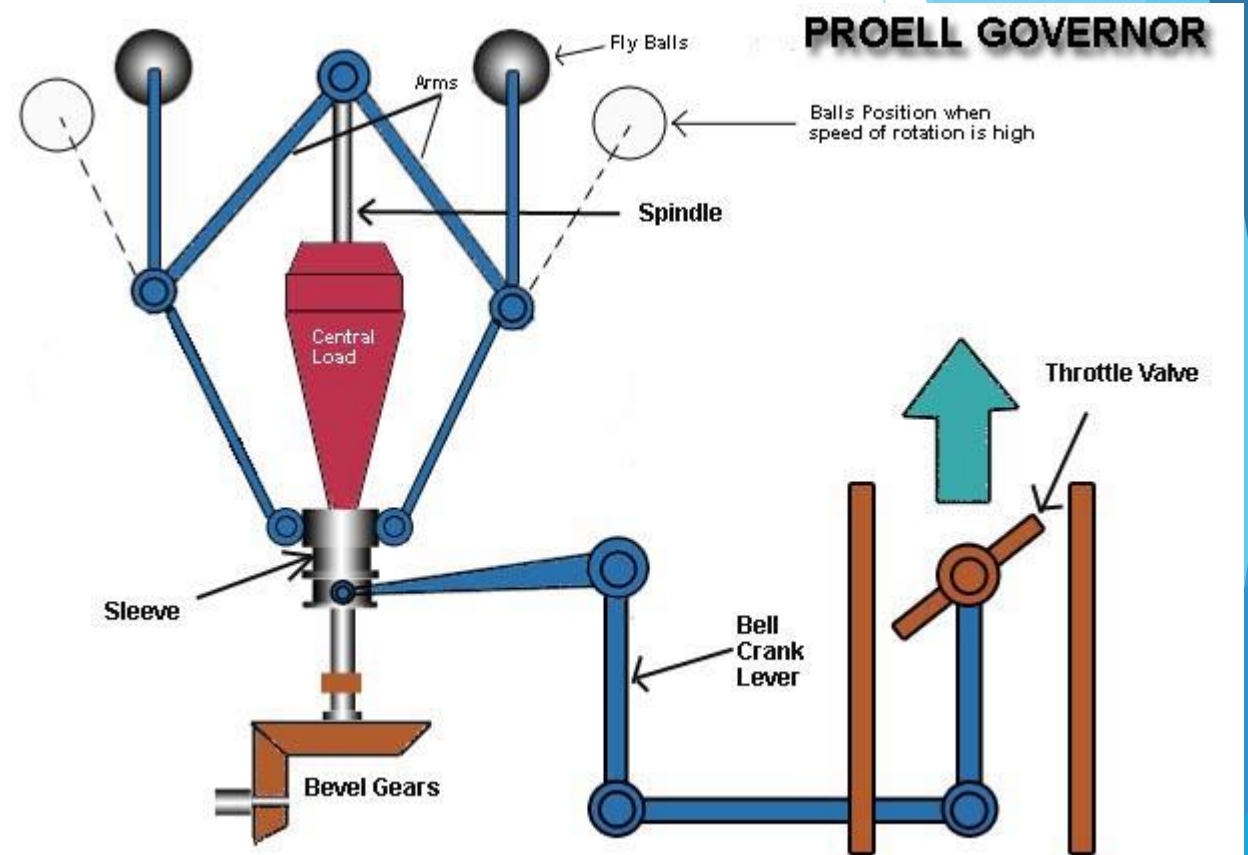


Fig. 1. Proell Governor



Gravity Loaded: Proell Governor contd...

m = Mass of **each ball** in kg

w = Weight of **each ball** in N

M = **Mass of the central load** in kg

W = **Weight of the central load** in N

r = **Radius of rotation** in m

h = **Height of governor** in m

N = **Speed of the balls** in rpm.

ω = Angular speed of the balls $(2\pi N/60)$ rad/s

F_C = Centrifugal force acting on the ball N

T_1 = Force in the arm in N

T_2 = Force in the link in N

α = Angle of inclination of the arm to the vertical

β = Angle of inclination of the link to the vertical

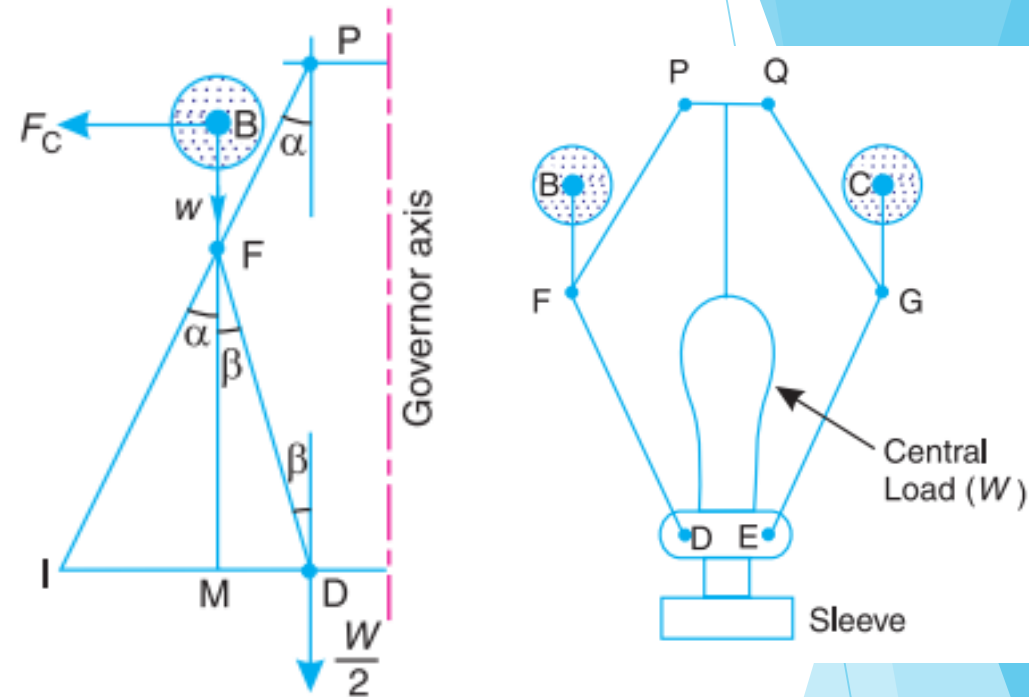


Fig. 2. Line diagram of Proell Governor

Proell Governor contd...

- Considered the equilibrium of the forces on one-half of the governor as shown in Fig. 3.
- **The instantaneous centre (*I*)** lies on the intersection of the line *PF* produced and the line from *D*.
- Drawn perpendicular to the spindle axis. The perpendicular *BM* is drawn on *ID*.
- Taking moments about instantaneous centre *I*:

$$F_c \times BM = w \times IM + \frac{W}{2} \times ID \quad (1)$$

$$F_c = mg \times \frac{IM}{BM} + \frac{Mg}{2} \times \frac{(IM + MD)}{BM}$$

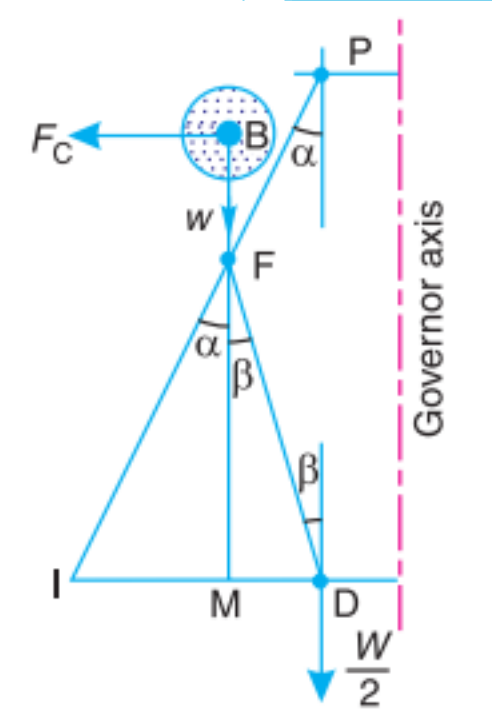


Fig. 3. Line diagram



Proell Governor contd...

Multiplying and dividing by FM , we get:

$$F_c = \frac{FM}{BM} \left[m.g \times \frac{IM}{FM} + \frac{M.g}{2} \times \left(\frac{IM}{FM} + \frac{MD}{FM} \right) \right]$$

$$F_c = \frac{FM}{BM} \left[m.g \times \tan \alpha + \frac{M.g}{2} \times (\tan \alpha + \tan \beta) \right]$$

$$F_c = \frac{FM}{BM} \times \tan \alpha \left[m.g + \frac{M.g}{2} \times \left(1 + \frac{\tan \beta}{\tan \alpha} \right) \right]$$

We know that, $F_c = m\omega^2 r$, $\tan \alpha = r/h$, $\tan \beta / \tan \alpha = q$

$$m.\omega^2.r = \frac{FM}{BM} \times \frac{r}{h} \left[m.g + \frac{M.g}{2} \times (1 + q) \right] \quad (2)$$

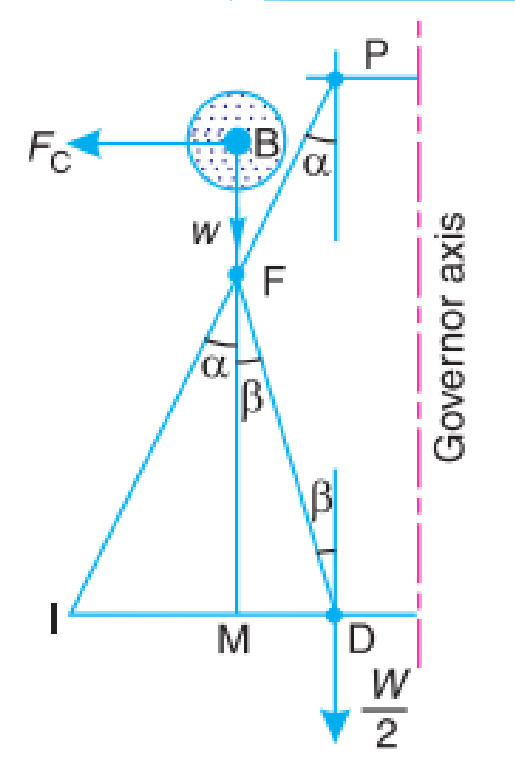


Fig. 4.



Proell Governor contd...

$$m.\omega^2.r = \frac{FM}{BM} \times \frac{r}{h} \left[m.g + \frac{M.g}{2} \times (1+q) \right]$$

$$h = \frac{FM}{BM} \times \frac{g}{\omega^2} \left[m + \frac{M}{2} \times (1+q) \right]$$

$$h = \frac{FM}{BM} \times \left[m + \frac{M}{2} \times (1+q) \right] \times \frac{895}{N^2} \quad (3)$$

When **the length of arms are equal to the length of links** and the points P and D lie on the same vertical line, then **$\tan\alpha = \tan\beta$ and $q = 1$**

$$h = \frac{FM}{BM} \times \left[\frac{m+M}{m} \right] \times \frac{895}{N^2} \quad (4)$$



Proell Governor contd...

When the loaded sleeve moves up and down the spindle; the frictional force acts on it in a direction opposite to that of the motion of sleeve.

$$h = \frac{FM}{BM} \times \left[\frac{2mg + (Mg \pm F)(1+q)}{2mg} \right] \frac{895}{N^2} \quad (5)$$

The + sign is used when the sleeve moves upwards or the governor speed increases and - sign is used when the sleeve moves downwards or the governor speed decreases.

Case 1: $q = 1$

$$h = \frac{FM}{BM} \times \left[\frac{mg + (Mg \pm F)}{mg} \right] \frac{895}{N^2}$$

Case 2: $F = 0$

$$h = \frac{FM}{BM} \times \left[\frac{2m + M(1+q)}{2m} \right] \frac{895}{N^2}$$

Case 3: $q = 1, F = 0$

$$h = \frac{FM}{BM} \times \left[\frac{m + M}{m} \right] \frac{895}{N^2}$$



Problem 1. In a Proell governor the mass of each ball is 8 kg and the mass of the sleeve is 120 kg. Each arm is 180 mm long. The length of extension of lower arms to which the balls are attached is 80 mm. The distance of pivots of arms from axis of rotation is 30 mm and radius of rotation of the balls is 160 mm when the arms are inclined at 40° to the axis of rotation. Determine the:

1. Equilibrium speed
2. Coefficient of insensitiveness if the friction of the mechanism is equivalent to 30 N
3. Range of speed when the governor is inoperative

Solution. Given : $m = 8 \text{ kg}$; $M = 120 \text{ kg}$; $r' = 160 \text{ mm} = 0.16 \text{ m}$;
 $b = c = 180 \sin 40^\circ = 115.7 \text{ mm}$; $r = b + 30 = 115.7 + 30 = 145.7 \text{ mm}$
 $a = 180 \cos 40^\circ = 137.9 \text{ mm}$, $EF = r' - r = 160 - 145.7 = 14.3 \text{ mm}$

$$BE = \sqrt{BF^2 - EF^2} = \sqrt{80^2 - 14.3^2} = 78.7mm$$

$$e = a + BE = 137.9 + 78.7 = 216.6 \text{ mm}$$

(i) Taking moment about I ,

$$mr\omega^2 \times e = mg \times (c + r - r') + \frac{Mg}{2} \times (b + c)$$

$$8 \times 0.16 \times \omega^2 \times 0.2166 = 8 \times 9.81 \times (0.1157 + 0.1457 - 0.16) + \frac{120 \times 9.81}{2} \times (0.1157 + 0.1157)$$

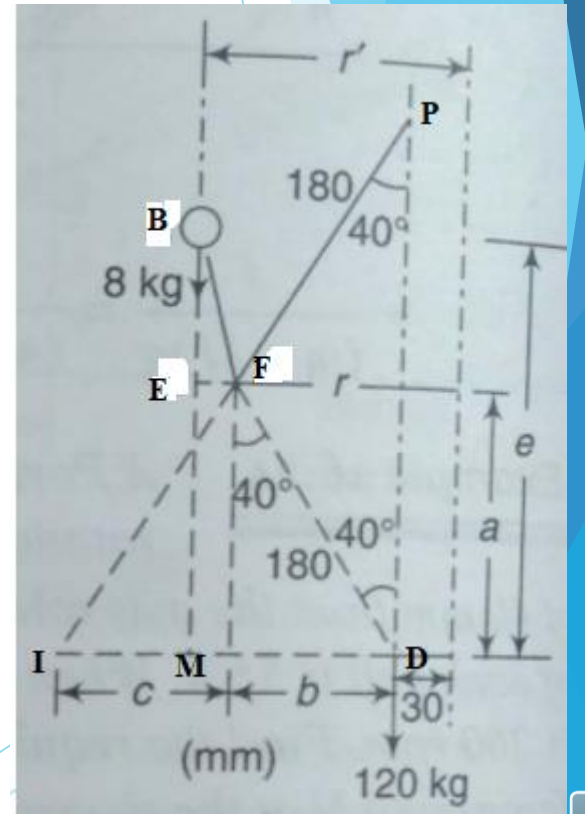


Fig. 5.



Problem 1. contd...

$$\omega^2 = 519.9$$

$$\omega = 22.8 \text{ or } N = \frac{22.8 \times 30}{\pi} = 217.7 \text{ rpm}$$

2. Considering the friction, let ω_1 and ω_2 be the maximum and minimum speeds respectively.

$$8 \times 0.16 \times \omega_2^2 \times 0.2166 = 8 \times 9.81 \times (0.1157 + 0.1457 - 0.16) + \frac{120 \times 9.81 - 30}{2} \times (0.1157 + 0.1157)$$

$$\omega_2^2 = 507.4$$

$$\omega_2 = 22.52 \text{ or } N_2 = \frac{22.52 \times 30}{\pi} = 215.1 \text{ rpm}$$

$$8 \times 0.16 \times \omega_1^2 \times 0.2166 = 8 \times 9.81 \times (0.1157 + 0.1457 - 0.16) + \frac{120 \times 9.81 + 30}{2} \times (0.1157 + 0.1157)$$

$$\omega_1^2 = 532.4$$



Problem 1. contd...

$$\omega_1 = 23.07 \text{ or } N_1 = \frac{23.07 \times 30}{\pi} = 220.3 \text{ rpm}$$

$$\text{Coefficient of insensitiveness} = \frac{N_1 - N_2}{N} = \frac{220.3 - 215.1}{217.7} \times 100 = 2.39\%$$

3. Range of speed = 220.3-215.1 = 5.2 rpm



Thank You

